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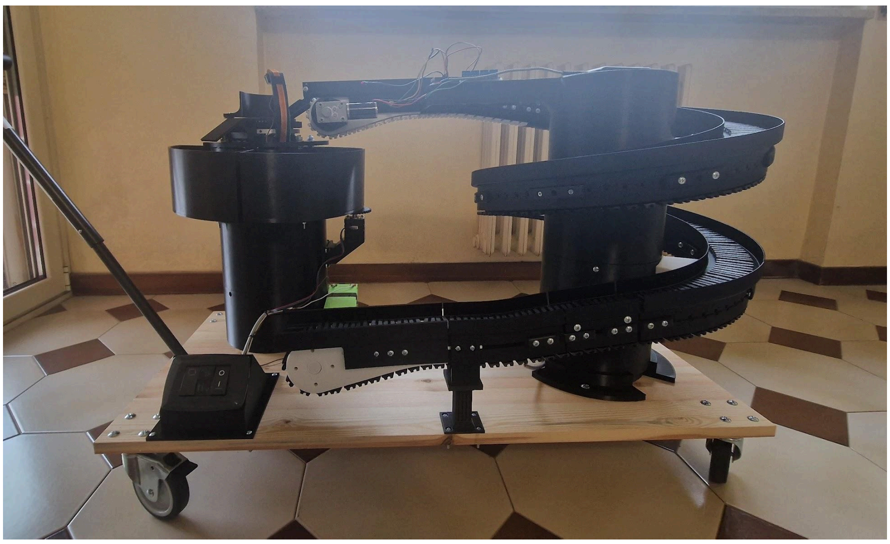
FUTURE INNOVATORS

SENIOR

Liceo Scientifico "E. Amaldi" Bitetto

ITALY



Halve the time,double the results

Argo is a smart system created to help archaeologists. It’s the first of its kind to be able to digitalize archaeological fragments via scanning, create 3D models of the pieces and categorize them systematically, doing all of this quickly and on-site.

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Team presentation

The team is composed by two senior students and one junior student from the "Edoardo Amaldi" Scientific High School in Bitetto, in the south of Italy: Elisabetta Liliana Albanese and Rossella Fumai who attend Applied Sciences scientific course and Simone Gismondi who attends the traditional scientific course. Our journey into robotics began a few years ago, thanks to school initiatives which led us to enjoy computer science, mechanics and electronics.

Figure 1: Our team

Liliana served as the project manager and took care of everything related to the communication of our project, from the documentation and presentation.

Simone was responsible for the **hardware** engineering and 3D design of the prototype, ensuring the physical structure was functional and optimized.

Rossella developed the software and engineered the Artificial Intelligence. She was also responsible for the graphics of our project.

Summary of the project idea

The Italian archaeological landscape represents an invaluable scientific and cultural resource, with an estimated five million artifacts currently stored in the repositories of state museums. However only approximately **25%** of state-owned heritage is actually **accessible to the public**, while the remaining **75% remains stored** for periods often exceeding fifty years. This critical situation is not attributable to a lack of institutional interest, but rather to a huge **human and physical limit** inherent in post-excavation analysis methodologies. While stratigraphic extraction techniques have modernized, the subsequent phases of **cleaning, cataloging, and manual reconstruction** still rely on purely analog workflows. The production of a single ministerial catalog entry for a ceramic lot requires an average of **20 to 40 hours of specialized labor**. This leads to millions of fragments being archived and forgotten for a long time.

In response to this challenge, **Argo** has been developed as an advanced technological ecosystem designed to optimize the archaeological workflow through immediate **digitalization and systematic categorization**. Argo is the first integrated system of its kind capable of generating high-resolution 3D models of fragments and classifying them directly at the excavation site. The primary objective is the **drastic reduction of processing times**, transforming procedures that currently last years into **real-time** operations.

The Open Source nature of the project is the fundamental pillar for overcoming the fragmentation of global heritage. By making it freely available online, we want to facilitate **sharing digital data across a global network.** A significant example of this necessity is the 2022 discovery of the marble head of Hercules at the Antikythera shipwreck site, which was identified as belonging to a torso recovered over a century earlier, in 1900. Without a shared digital platform, the reunification of elements discovered across different centuries and geographical contexts remains a **rare and arduous event**. Argo could enable archeologists to consult a catalogue of pieces found in a specific place or with really similar characteristics to the one they are searching for.

Consequently, the adoption of Argo ensures not only an **increase in scientific productivity** but also an improvement in global historical understanding.

Robotic solution

The idea

Thinking about the theme of the competition, we wondered which problems were still unsolved in such an expanding sector. Looking at museums, we asked ourselves why we are so used to seeing the **same artifacts** over and over, and why it is so rare for new finds to be added. This led us to focus on **archaeology**, a vast and complex subject that we began to study more closely.

By talking to experts and conducting our own research, we realized that one of the biggest issues is the **time** required for the **analysis and categorization** of pieces. Quite unexpectedly, we learned that once a fragment is found, it might have to **sit in a container for several months** before being cleaned, analyzed, and categorized. This happens because the speed of modern excavations is far higher than the speed at which the finds can be studied.

To solve this problem, we went through several attempts to find what fit better our goal: create a project that could **support and simplify** the work of archaeologists, **not replace them**. This is how we came up with Argo, a system designed to speed up manual tasks in a smart and efficient way, while being versatile enough to **work directly at the excavation site**.

At the moment, there is no other system quite like ours, and the archaeologists we spoke to find it incredibly useful. However, we are aware that this is just the **beginning** and that we still have a long way to go.

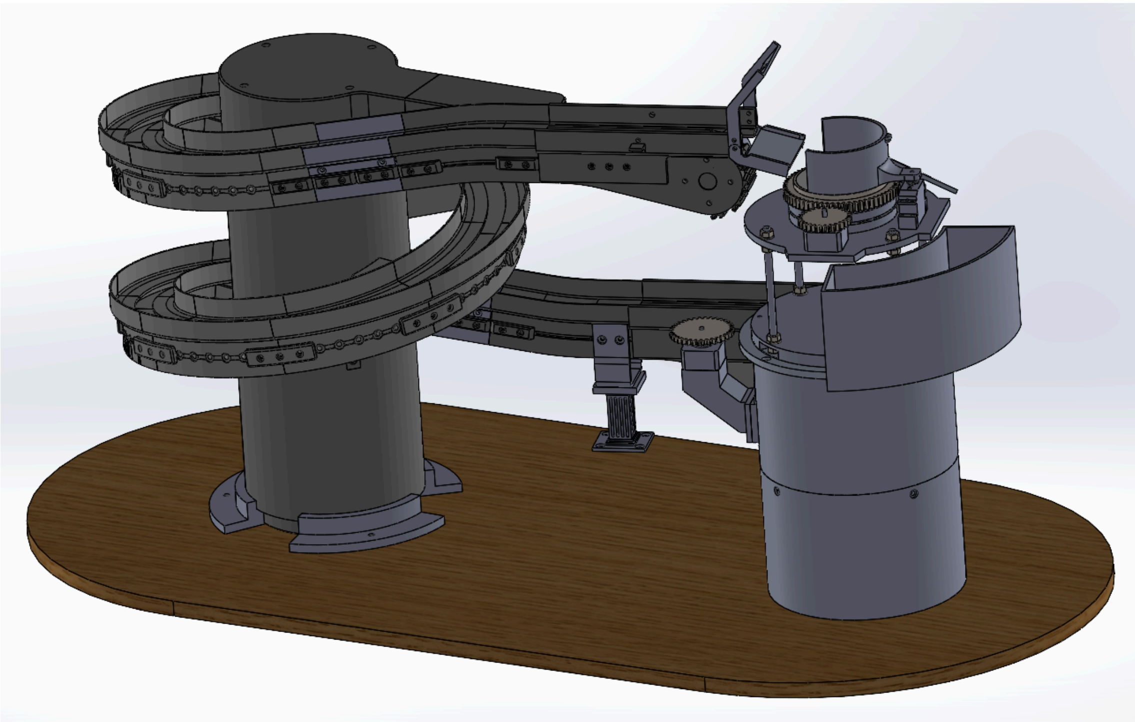

Figure 2: 3D model

General workflow of the prototype

The workflow of the Argo prototype follows a **smooth, automated cycle** designed to make field work **simple and effective**. Once the system is powered on, fragments are loaded one by one into the helix conveyor, which acts as a **storage buffer** and regulates the flow of pieces. From there, each find is transported toward the heart of the device: **the scanner**.

As soon as the fragment reaches the correct position, the conveyor pauses to allow for a **precise 360-degree acquisition**; during this stage, the secondary PiCamera moves along its vertical arched rack to capture the artifact from **multiple elevations** while the system analyzes the object to identify its **category**. Simultaneously, the bins at the base of the scanner **rotate** to align according to the detected category. At this point, a mechanical paddle moves the piece gently into the corresponding container. Once the sorting is complete, the conveyor resumes its movement, ready to process the next fragment in a continuous and optimized cycle.

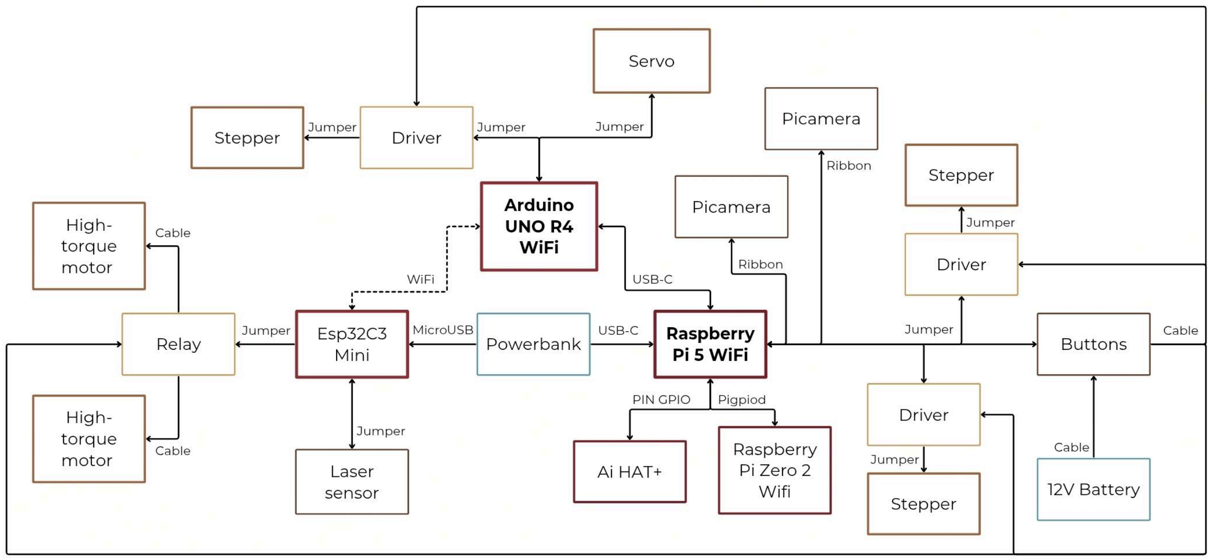

Figure 3: the workflow

Technical overview

The technical architecture of Argo is made of an integrated hardware and software ecosystem designed to **automate and optimize** the categorization and analysis of archaeological finds.

At the center of the operation there is a **Raspberry Pi 5** equipped with an **AI HAT+**. The Raspberry Pi acts as the **primary processing hub**, managing the system logic, including the activation, deactivation, and interruption of the scan process via external buttons. It is directly responsible for the Scanner Module, where it orchestrates **two PiCameras** and utilizes a **Raspberry Pi Zero 2 WiFi as a remote factory** to drive **two stepper motors** to generate comprehensive digital models of each artifact.

The mechanical flow is synchronized through an **Arduino** unit that serves as a vital communication bridge between the Raspberry Pi and the rest of the peripheral hardware. The Arduino manages the **Sorting Bin motor** to ensure fragments are correctly categorized post-scan. Furthermore, it maintains a wireless link via Wi-Fi with an **ESP module** located at the upper end of the conveyor belt.

The detection sequence is triggered by a **laser sensor** integrated with the ESP. When a fragment is detected passing into the scanner zone, the ESP communicates the status to the Arduino and immediately **halts the conveyor motors**, meanwhile the Arduino tells the Raspberry Pi to start the scanning process. As the first photo is captured by the PiCamera, the **AI HAT+ immediately analyzes the image to classify the artifact**; the Raspberry Pi then communicates this data to the Arduino, instructing it to move the Sorting Bins to the correct position. Once the scan is finalized, the system resumes operation, moving the piece into its designated bin.

Mechanics

The system is structured into **four main units**, each designed to perform a specific function within the overall architecture. The entire framework is composed of both 3D-printed components and specialized mechanical parts, having **ABS** as the primary material. This thermoplastic polymer was chosen for its excellent compromise between **structural resistance**, **cost-efficiency**, and **ease of processing**, ensuring the device is durable enough for the rigorous conditions of an active excavation site.

Helix conveyor

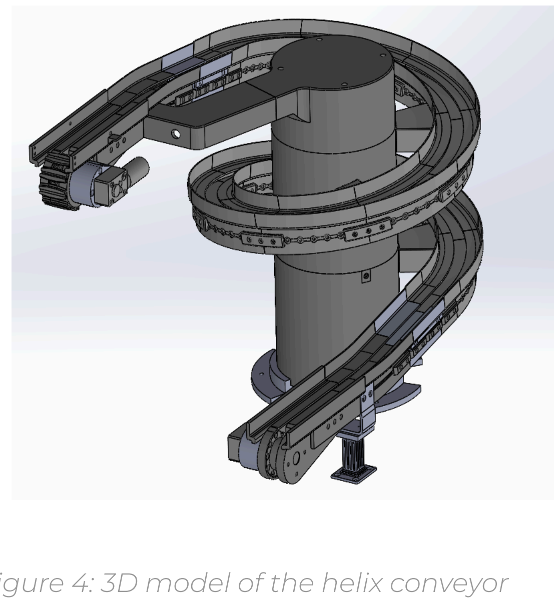

Figure 4: 3D model of the helix conveyor

This unit consists of a central cylindrical core around which a helical ramp is secured to support and guide the conveyor belt. Measuring 530 mm in height and 795 mm in length, the **conveyor is 3D-printed in ABS** to maintain a lightweight yet sturdy profile. The motion is driven by **two high-torque and self-locking motors** controlled by an **ESP32C3 Mini** mounted on top of the structure. The conveyor belt remains in operation until an object is detected by a **laser sensor** positioned at the last tract of the ramp. Once the laser beam is interrupted, the ESP32 immediately **halts the motors**, transitioning the unit into a **buffer mode**. This allows the system to hold the artifacts in a secure position before they enter the scanning phase, preventing congestion and ensuring that each piece is handled with the necessary care to avoid damage.

Scanner module

This unit is positioned atop a supporting tower, serving as the dedicated **core station for high-resolution 3D mapping**. The module is engineered around a central **transparent acrylic turntable** measuring 10 cm in diameter, which rests upon a **6-inch slewing bearing**. Rotation of this turntable is achieved through a **robust gear-to-gear transmission system** driven by a dedicated **stepper motor** mounted securely on the side of the module's base plate. Directly beneath the transparent surface, aligned with a central aperture, the first **PiCamera** is fixed to capture detailed underside views. To manage image contrast for AI analysis, the scanning area features an integrated **neutral background**; this consists of a paddle within the curved scanning shield, providing a consistent backdrop. **Dynamic capture from multiple angles** is handled by a **large arched gear** that passes through the turntable at two distinct points. This entire arched structure is driven by an **additional stepper motor** located beneath the main plate, acting through a **secondary gear transmission**. One of the two ends of this arch features a place designed to attach the **second PiCamera**, allowing it to move precisely along an arc. The synergy between the rotating turntable and the orbital movement of the camera ensures a complete and precise **360-degree digital mapping**, providing all the necessary data for affinity analysis.

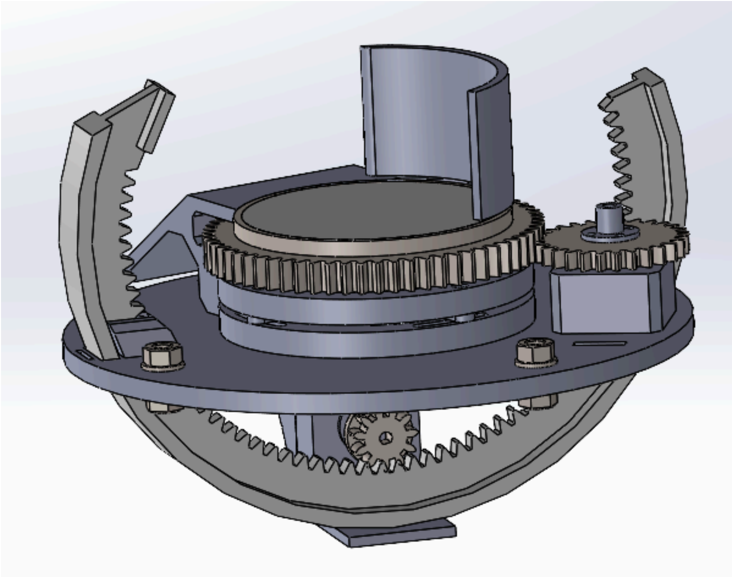

Figure 5: 3D model of the scanner station

Sorting unit

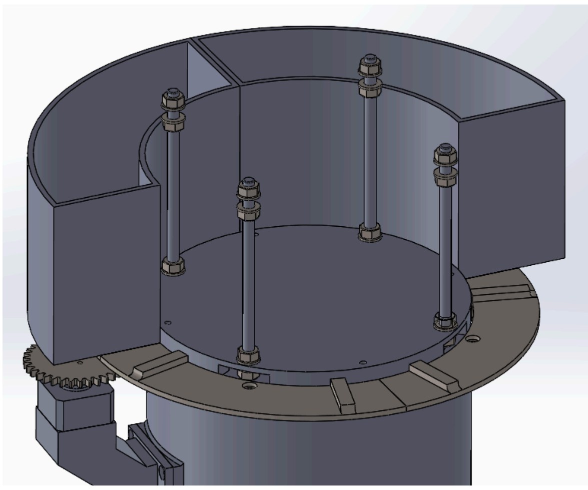

Figure 6: 3D model of the sorting unit

The Sorting Unit is integrated directly beneath the scanning station. To maintain a compact and efficient footprint, the **scanner module is suspended** by a series of brackets, creating an open area covered by the **collection bins**. At the core of this sub-structure, the system's primary electronics, including the **Arduino, the Raspberry Pi 5 with its AI HAT+, the Raspberry Pi Zero 2 Wifi, and three motor drivers**, are securely mounted. The sorting bins themselves are housed on a **specialized interlocking geared platform** designed for **rapid removal**, allowing archaeologists to easily swap or empty them during field operations. This entire circular assembly is mounted on a slewing bearing and is rotated by a **side-mounted stepper motor** that aligns the correct bin with the output path based on the AI's classification. To finalize the cycle, the system utilizes the **servo-driven paddle** mentioned in the scanning phase; once the analysis is complete, this paddle acts as a **mechanical arm** to gently push the artifact from the acrylic turntable into the designated bin below. This seamless integration ensures that each find is not only digitally archived but also physically categorized with minimal manual intervention.

Mobile Framework & Control Interface

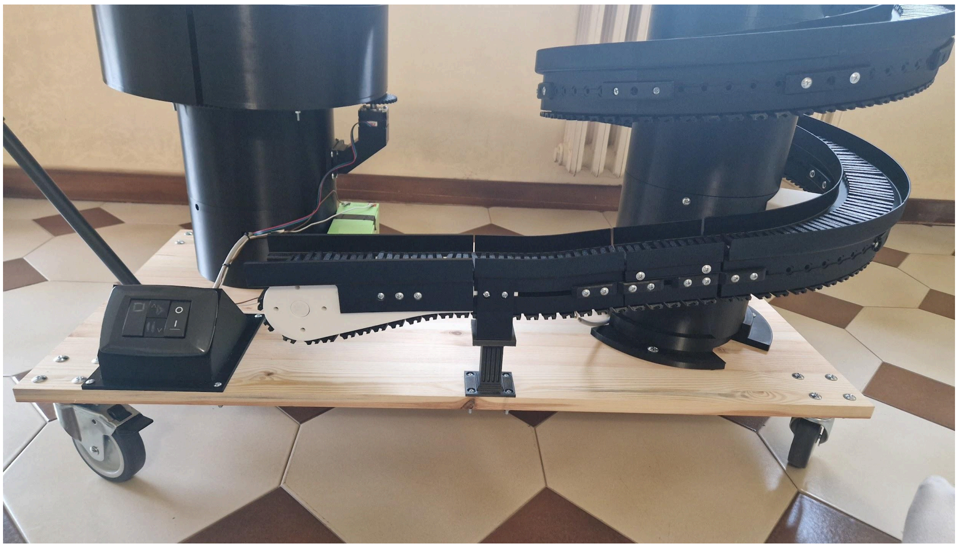

Figure 7: Framework & control interface

The structural integrity of Argo is consolidated in a **mobile base unit** that rests on a solid 18 mm thick multilayer wood platform, with overall dimensions of 1 m x 0.5 m, which provides a stable foundation for the vertical components. To ensure maneuverability across excavation sites, the base is equipped with **four heavy-duty swiveling wheels**, featuring two standard models and two with integrated safety locking brakes to maintain stability during scanning phases. While not present in the current prototype, the final production version is designed to be encased in a **full aluminum housing with a matte black internal coating**, specifically engineered to shield the scanning environment from external light interference and **protect internal components from environmental factors**.

Operational workflow is managed via a **remote control unit** connected to the main chassis by a multi-core cable, housing three primary physical commands for immediate handling. This interface includes a momentary "push-and-return" interruption button for the **instant suspension of operations**, a dual-action toggle switch where the downward command **halts the scanner cycle** and the upward command triggers a **manual restart**, and a dedicated ON/OFF toggle for the **general power supply**. In the final version of the project the interface will integrate two multi-function status LEDs to provide **real-time diagnostic feedback**. The first LED **monitors the operational state**, transitioning from Blue during initialization to Green during conveyor motion, Yellow during scanning, and Red in the event of a system error. The second LED serves as a **power management indicator**, displaying Green for a full charge, Yellow for medium capacity, and Red when immediate charging is required.

Electronics

The electronic architecture of Argo is based on a distributed control system designed to manage high-speed AI processing, precise mechanical actuation, and real-time sensor feedback.

Raspberry Pi 5 with AI HAT+

This unit manages the **scanning process** and the transmission of captured images to the external computer. It interfaces with a **Raspberry Pi Zero 2 WiFi via remote GPIO protocols to coordinate the two stepper motors** responsible for rotating the transparent turntable and actuating the mobile camera arm. By leveraging the AI HAT+, the board provides the necessary **processing power for real-time categorization** during the acquisition cycle.

Arduino UNO R4 WiFi

This board is responsible for managing the **sorting mechanics**, specifically controlling the **stepper motor** that rotates the collection bins and the **servo motor** that operates the background/sorting paddle. It acts as a **bridge for system synchronization**, communicating via serial interface with the Raspberry Pi to monitor the scan status (start/end) and via Wi-Fi with the ESP32 to coordinate the conveyor's operation, receiving "halt" signals and transmitting "restart" commands.

ESP32 + Laser Tripwire

This module is interfaced with a **laser sensor system**, consisting of a laser emitter and a photocell positioned on opposite sides of the helix conveyor. The ESP32 detects when an object passes through the beam, immediately **halting the conveyor motors** to position the fragment for scanning. It then communicates the motor status wirelessly to the Arduino to ensure **synchronized workflow transitions**.

PiCamera

The imaging system utilizes two picameras for full-artifact documentation. The first PiCamera is positioned beneath the transparent acrylic turntable to acquire detailed **photos of the piece's underside while it rotates**. The second camera is mounted on the large arched gear that passes through the turntable at two distinct points; this camera performs **dynamic acquisition**, in a full rotation of the turntable the camera moves upward by 45 degrees to capture the subject from multiple elevations.

Control Buttons

The physical interface consists of **three specialized switches** housed within a remote control unit, connected to the **main framework by a multi-core cable**. This setup includes a momentary "push-and-return" interruption button for the instant **suspension of operations** and a dual-action toggle switch, where the downward command **halts the scanner cycle** and the upward command **triggers a manual restart**, both of which are interfaced via GPIO with the Raspberry Pi. A third dedicated ON/OFF toggle serves as the **general power supply switch**, directly connected to the battery system.

Battery System

The prototype is powered by a **12V AGM (lead-acid) battery**, which provides the necessary current for the helix conveyor and the stepper motors. **For the demonstration phase**, the logic boards (Raspberry Pi 5 Wifi, Raspberry Pi Zero Wifi, Arduino, ESP32) and the servo motor are powered by a **dedicated power bank**, ensuring stable voltage regulation and preventing interference between the high-draw mechanical components and the sensitive electronic circuitry.

Software

The system's intelligence is distributed across **four hardware platforms**, utilizing a **multi-language programming environment**. The Raspberry Pi 5 and the Raspberry Pi Zero 2 Wifi operate on the **Debian Bookworm** distribution, paired with the **Hailo software suite** specifically for the AI HAT+ integration. The Arduino UNO R4 WiFi and ESP32C3-mini are programmed using the **Arduino IDE framework**. Python serves as the primary language for high-level logic on the Raspberry Pi and for the code running on the PC-server, while C++ is utilized for the real-time firmware execution on the microcontrollers.

Raspberry Pi 5 and AI Hat+

The high-level software architecture comprises a main control script and a dedicated sub-module for the scanning process, both written in Python and **executed via the command-line interface (CMD)**. The main script manages the **overall system synchronization**, including the communication bridge with the Arduino, the handling of operational pauses, and the AI-driven classification of fragments. The **scanning sub-module** contains the logic for remote stepper motor actuation **via the Pi Zero 2 remote factory**, image acquisition, and the subsequent data management, allowing for files to be **transmitted to an external computer** or **saved directly to an external USB drive**.

Arduino UNO R4 WiFi

The Arduino firmware serves as the **communication gateway** between the ESP32 and the Raspberry Pi. It operates in "listening mode" until the ESP32 signals a conveyor halt; it then transitions to an active state to notify the Raspberry Pi to begin the scan and, once complete, informs the ESP32 that the conveyor cycle can resume. Simultaneously, the board **manages the mechanical sorting logic**, positioning the designated collection bin based on the classification results and actuating the servo-driven paddle to slide the fragment off the turntable.

Esp32C3-mini

The ESP32 software focuses on real-time sensor monitoring and conveyor control. It constantly polls the **laser tripwire** system to detect the arrival of an artifact at the top of the ramp. Upon detection, the code triggers an immediate halt of the conveyor motors and **transmits a status update to the Arduino via a wireless URL request (WiFi)**, ensuring a seamless transition between the transport and scanning phases.

Scanning process and elaboration

The workflow begins when the Arduino receives a wireless signal from the ESP32, subsequently notifying the Raspberry Pi via serial communication to **initiate the scanning sequence**. Using the initial frame captured, the AI HAT+ processes the image through a **custom-trained neural network model to identify the artifact's type and category**. This classification is immediately transmitted back to the Arduino, which **rotates the sorting assembly to align the corresponding collection bin**.
While the sorting system prepares the hardware, the Raspberry Pi executes the **primary scanning script**. The process involves a rotation of 900° (2.5 full turns), driven by the Raspberry Pi Zero 2 under remote command, during which both the fixed lower PiCamera and the mobile arched PiCamera capture a synchronized photograph every 10°. These images are automatically archived in a directory timestamped with the date and hour of the scan, subdivided into dedicated folders for each camera's perspective. Concurrently, the **system generates a metadata file** containing the artifact's discovery data, stored directly within the target 3D directory.

Upon completion of the rotation cycle (approximately 4 minutes), the Raspberry Pi transfers the image folders to the server-PC into their respective directories (e.g., "pi_receive" and "3d files"), accompanied by an empty "done.txt" trigger file. This file serves as a **signal for the workstation to begin the reconstruction phase**. Once the data transfer is verified, the Raspberry Pi signals the Arduino to actuate the servo-driven paddle, which gently **guides the artifact down a ramp into its designated bin**.

On the server side, a Python script continuously monitors the reception directory for the "done.txt" file. Once detected, the workstation identifies the most recent folder and initiates the **photogrammetry pipeline** via command line. The reconstruction is performed using the **OpenMVG C++ library to generate a high-density point cloud**, followed by **OpenMVS to transform the spatial data into a textured 3D solid**. The resulting 3D model is then saved in the "3d files" repository, consolidated with the previously generated artifact metadata.

Challenges faced

Working on this project, we were able to grow and learn a lot. We had to face many challenges, beginning with the conceptual foundation of the idea itself. In the first place, we struggled to find a solution that was highly efficient while ensuring it would **support, rather than replace**, the vital work of archaeologists. Our goal was to create an augmented workflow where the system handles the **repetitive manual labor**, allowing the **human expert to focus entirely on high-level scientific interpretation and historical synthesis**.

Another significant hurdle was the technical classification of the artifacts. Deciding how to divide the finds into distinct categories was a complex task due to the **sheer diversity of materials found on a dig**. To solve this, we sought the expertise of professional archaeologists, whose guidance was crucial in defining the system's sorting logic. Thanks to their input, we established four specific bins to categorize items such as **plain pottery, decorated ceramics, organic remains (including bones and shells) and metals**, ensuring that Argo's output aligns perfectly with real-world museum and archival standards.

Finally, we addressed the critical issue of artifact preservation. Dealing with ancient, often brittle materials meant that every stage of Argo's movement had to be **engineered for maximum safety**. We spent a significant amount of time refining the physical transitions within the prototype to **prevent any risk of breakage**. By integrating a series of **specialized ramps**, we managed to soften every potential impact, ensuring that the fragments are handled with the utmost care as they move from the helix conveyor through the scanner and into their final designated bins. This focus on physical integrity ensures that Argo is not just fast, but also a **safe and reliable tool for preserving our heritage**.

Cost of the prototype

Total cost of the prototype: € 560.5

Boards: € 224
- Raspberry Pi 5 WiFi: € 90
- AI HAT+: € 75
- Arduino UNO R4 WiFi: € 30.50
- Esp32C4 Mini: € 8.50
- Raspberry Pi Zero 2 WiFi: € 20

10 kg ABS filament : €120

Mechanical parts: € 40
- Slewing bearing (6 inch): € 15
- Slewing bearing (14 inch): € 25

Motors: € 76
- 3 Stepper motors : € 45 (€ 15 each)
- 2 High-torque motors: € 27
- Servo motor: € 4

Electronics: € 31
- 2 Picameras: € 8
- Buttons: €2
- 3 Stepper drivers : €17 (€5.6 each)
- Laser sensor: € 4

Additional materials: € 69.5
- M3 and M4 Threaded inserts: € 4
- Screws + Bolts + Washers : € 15
- Brackets: € 2
- Bearings: € 3.5
- Wooden table : € 10
- Wheels: € 18
- Acrylic turntable: € 17

It is important to note that the current prototype cost of approximately €600 reflects **retail prices for individual components and one-off custom parts**. In a mass-production scenario, the **unit cost would be significantly reduced** through bulk procurement of electronics and industrial manufacturing processes. Furthermore, within the professional archaeology sector, this figure represents an exceptionally **low investment**. Compared to the high costs of traditional survey equipment and the hundreds of man-hours typically required for manual cataloging, Argo offers a **high-performance, cost-effective solution** that democratizes access to advanced digital documentation and AI analysis.

Social impact and innovation

Social impact

As previously stated, the Italian and international archaeological landscape is currently dominated by a structural paradox: the technical capacity to extract artifacts from the ground has far outpaced the human capacity to study, catalog, and exhibit them. This asymmetry has generated a systemic crisis in museum repositories.

Estimates based on the Ministry of Culture and ISTAT censuses reveal that approximately **75%** of Italy’s archaeological heritage, totaling over **five million artifacts**, is trapped in crates within warehouses inaccessible to the public. Many of these fragments remain stored for decades, losing their educational potential. The root of this paralysis does not lie in a lack of funding, but in the physical time required by official cataloging standards. Following the directives of the ICCD (Central Institute for Catalogue and Documentation), the analog process for a single ceramic fragment involves washing, drying, manual labeling with ink, technical drawing, photographic documentation, and the compilation of the official "TMA" (Archaeological Materials Table) sheet. This entire cycle takes an average of **20 to 45 minutes per individual piece**. Consequently, processing a modest crate containing a hundred fragments requires **over a week of highly specialized labor**.

In this context, the social impact of **Argo** manifests as a technological and logistical revolution. By reducing cataloging time to **less than 5 minutes** per fragment through 3D scanning and artificial intelligence, Argo transforms archaeology from an accumulation discipline into a science of real-time sharing. This paradigm shift allows local museums to convert finds into temporary exhibitions or permanent collections in record time, revitalizing public interest and decentralizing tourism toward internal areas that currently suffer from marginalization.

However, Argo’s true potential is expressed on a global scale through the creation of a collective **Open Source archive** aimed at researchers worldwide. Until now, solving complex historical puzzles has depended on the visual memory of individual archaeologists or lucky coincidences. A striking example is the 2022 discovery of the marble head of Hercules at the Antikythera shipwreck site in Greece; the piece was linked to a torso kept in Athens that had been recovered over a century earlier, in 1900. While exceptional, this event highlights the absurdity of relying on chance to reconstruct human history.

Innovation

The core innovation of Argo lies in its ability to **bridge the gap between physical excavation and digital preservation** through a fully integrated, automated ecosystem that operates directly on the front lines of discovery. Unlike traditional methods that rely on fragmented workflows, Argo represents the first system of its kind capable of performing high-resolution 3D scanning, systematic categorization, and digital archiving in a single, streamlined process.

The technological breakthrough centers on the synergy between its specialized hardware, featuring a helix conveyor for precise fragment handling, and an AI-driven software core that identifies and catalogs pieces in real-time. By bringing the laboratory directly to the trench, Argo eliminates the need for transporting artifacts to remote facilities, thereby removing the logistical barriers that have historically kept 75% of our heritage hidden in storage. This is not a tool designed to replace the archaeologist; rather, it is a sophisticated collaborator engineered to handle the **most repetitive and time-consuming manual tasks** with unprecedented speed and efficiency.

The result is a transformative leap in the field: a platform that generates an immediate, Open Source digital archive as soon as a fragment touches the light, allowing professionals to focus their expertise on historical interpretation while the system ensures that no piece of history is ever lost to the silence of a warehouse again.

Entrepreneurship aspects

Throughout these months, we have honed our technical expertise to elevate Argo into a **fully functional system**. Now, as we transition from a field-tested prototype toward a market-ready industrial product, we recognize that the path to the industry involves demands that go beyond technical capabilities alone. To scale the project effectively, we must focus on **industrialization** and **mobility**, transforming our current setup into a ruggedized, **mass-producible unit capable of handling the harsh conditions of international excavation sites**.

To realize this vision, we have identified specific resources and professional skills essential for the next phase of development. Specifically, we require expertise in **industrialized electronics and mechanical design** to ensure that the tracking systems and hardware components are **durable and reliable in diverse environments**. Given the complexity of integrating a mobile autonomous base with a high-precision AI-driven Affinity Engine, a **senior professional with a strong background in industrialized design** would be invaluable in guiding the transition from a laboratory prototype to a global commercial product. Furthermore, securing **financial resources and investment** is critical to support the research and development of our software upgrades and the manufacturing of the physical units. By focusing on these entrepreneurship aspects, we aim to make Argo not just an innovative idea, but a **scalable solution that can be deployed by archaeological institutions worldwide**.

Confrontation with experts

The development of Argo was fundamentally shaped by a series of **consultations with experts from both the archaeological and engineering sectors**, ensuring that the system meets real-world professional standards and field requirements.

We worked most closely with **Professor Schiavariello**, the **primary archaeological consultant for the project**, who provided continuous guidance in refining our core concepts. Through multiple meetings and feedback gathered from his professional network, he helped the team **identify high-impact features** while discarding redundant solutions. Professor Schiavariello considers Argo a breakthrough for the industry, specifically because it addresses the **sector's most pressing cataloging backlogs without displacing the professional role of the archaeologist**. Unlike competing projects that aim to replace human expertise with complex robotic manipulators, he noted that Argo serves as a **powerful diagnostic tool that augments the researcher's capabilities while preserving the essential analytical nature of the work**.

Further methodological insights were provided by **Professor Silvio Pice**, **Director of the Archaeological Museum of Bitonto**. By granting us direct access to historical artifacts and explaining the rigorous traditional processes required to study and document a find, he allowed the team to fully **adopt the perspective of a field professional**. Professor Pice particularly praised Argo's **on-site versatility**, confirming its potential for i**mmediate deployment during excavation campaigns to streamline the transition from discovery to digital archiving**.

Figure 8: Museum of Bitonto

Finally, the system's technical and mechanical feasibility was validated by **Engineer Falagario**, who provided a **critical engineering review of the project**. His expertise was instrumental in **optimizing the mechanical assemblies and the precision of the gear systems**, particularly regarding the synchronization of the arched scanner and the helix conveyor. By **evaluating the structural integrity and the reliability of the motorized components**, Engineer Falagario ensured that Argo's design is not only innovative but also **industrially robust and ready for the challenges of a rugged field environment**.

Next steps

Moving forward, we will evolve Argo into a **fully autonomous field assistant**, transforming it from a stationary device into a **dynamic** partner on the excavation site. Our primary focus is the integration of an advanced **smart tracking system**, allowing Argo to **follow archaeologists** across the rugged terrain of a dig. By positioning the unit just outside the active trench, researchers can scan and categorize fragments the moment they surface, ensuring no data is lost to environmental exposure. Beyond its analytical capabilities, the device will serve as an **intelligent storage hub**, facilitating the organized and safe transport of artifacts across the site.

The next frontier for Argo's software is the implementation of an **AI-driven Affinity Engine**. Utilizing the high-resolution 3D models generated during the initial scan, the system will calculate the **morphological and stylistic compatibility** between different fragments. By providing archaeologists with an "**Affinity Percentage**," Argo will suggest potential joins and guide the physical reconstruction of artifacts. This leap in technology will drastically reduce the **trial-and-error nature of restoration**, allowing for the rapid reunification of history with a level of precision that was previously impossible. Through these advancements, Argo will not only document the past but actively help rebuild it.

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